Certified Kubernetes Administrator (CKA)

By Cloud Mentor Pro

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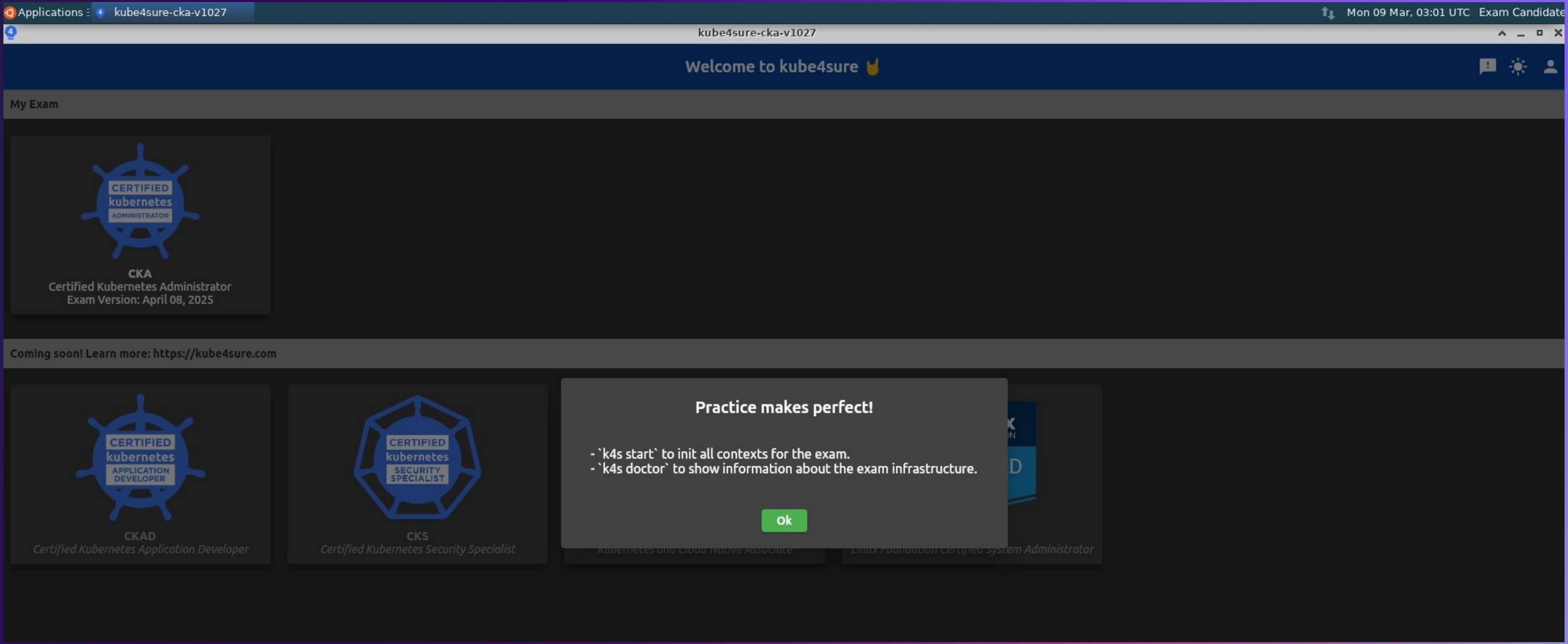
Kube4sure - Real CKA Exam (18 câu)


Applications
kube4sure-cka-v1027
Mon 09 Mar, 03:01 UTC
Exam Candidate
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CERTIFIED kubernetes APPLICATION DEVELOPER
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Practice makes perfect!
- `k4s start` to init all contexts for the exam.
- `k4s doctor` to show information about the exam infrastructure.
Ok
Linux Foundation Certified System Administrator

CKA-EXAM **Alex Pham**

Pre-Setup

Pre-Setup

Notes

Kube4Sure will automatically sets up cluster for you. Let's enjoy your exam!

Minimal setup

```
source <(kubectl completion bash)
echo "source <(kubectl completion bash)" >> ~/.bashrc
```

You can also use a shorthand alias for kubectl that also works with completion:

```
alias k=kubectl
complete -o default -F __start_kubectl k
```

Appending `--all-namespaces` happens frequently enough that you should be aware of the shorthand for `--all-namespaces` to `-A`

```
kubectl get all --all-namespaces
kubectl get all -A
```

The following settings will already be configured in your real exam environment in `~/.vimrc`. But it can never hurt to be able to type these down:

```
set tabstop=2
set expandtab
set shiftwidth=2
```

The expandtab make sure to use spaces for tabs. Memorize these and just type them down. You can't have any written notes with commands on your desktop etc.

Reference to https://kubernetes.io/docs/reference/kubectl/cheatsheet

Terminal 1 (Main)

```
student@kube4sure:~$ k4s start
Loading license successfully...
Please wait a moment to initialize the exam contexts....
Cluster ek8s is created...
Cluster mk8s is created...
Cluster k8s is created...
Provisioning all exam clusters...
All contexts are ready for the exam. Enjoy!
student@kube4sure:~$ k4s doctor
[OK] Internet is reachable.
[OK] chronyc is running.
[OK] Docker is running.
[03] Number of Kubernetes clusters: 03
[07] Number of running nodes: 07
[X] Pod 'big-corp-app' is not in Running state. Current status: Pending
[X] Pod 'foo' is not in Running state. Current status: Pending
[X] Pod 'front-end-858dc99445-mcbks' is not in Running state. Current status: Pending
[X] Pod 'front-end-858dc99445-zs2rh' is not in Running state. Current status: Pending
[X] Pod 'overloaded-cpu-6hxtd' is not in Running state. Current status: Pending
[X] Pod 'overloaded-cpu-lscbx' is not in Running state. Current status: Pending
[X] Pod 'overloaded-cpu-wbb2m' is not in Running state. Current status: Pending
[X] Pod 'presentation-5c4959c759-gdjf8' is not in Running state. Current status: Pending
[X] Pod 'presentation-5c4959c759-vhjsm' is not in Running state. Current status: Pending
[X] Pod 'web-0' is not in Running state. Current status: Pending
[X] Not all pods in k8s are running.
[OK] All pods in ek8s are running.
[OK] All pods in mk8s are running.
student@kube4sure:~$
```

📊 Tổng kết toàn bộ exam (18 câu)

Domain	Câu	Số câu
🔐 Security	Q1	1
⚙️ Cluster Maintenance	Q2 Q3 Q4	3
🌐 Networking	Q5 Q6 Q17	3
📦 Workloads & Scheduling	Q7 Q8 Q10 Q13 Q15	5
💾 Storage	Q11 Q16	2
🧩 Cluster Architecture	Q9 Q14 Q18	3

Q1. RBAC (Role-Based Access Control)

CKA-EXAM Alex Pham

1/18 Check

Question 1

Task weight: 3%

Set configuration context:

```
student@kube4sure:~$ kubectl config use-context k8s
```

Context:

You have been asked to create a new ClusterRole for a deployment pipeline and bind it to a specific ServiceAccount scoped to a specific namespace.

Task:

Create a new ClusterRole named deployment-clusterrole, which only allows to create the following resource types:

- Deployment
- StatefulSet
- DaemonSet

Create a new ServiceAccount named cicd-token in the existing namespace app-team1.

Bind the new ClusterRole deployment-clusterrole to the new ServiceAccount cicd-token, limited to the namespace app-team1.

Hint 💡

Create a new ClusterRole which can be used the kubectl create command. Then, create a new ServiceAccount and RoleBinding.

Solution 🔥

Use the kubectl create command to create new a ClusterRole

```
student@kube4sure:~$ kubectl create clusterrole deployment-clusterrole --ve
```

Create a new ServiceAccount

```
student@kube4sure:~$ kubectl create serviceaccount cicd-token --namespace
```

Create a new RoleBinding

```
student@kube4sure:~$ kubectl create rolebinding deployment-binding --clust
```

Verify it's created correctly

```
student@kube4sure:~$ kubectl auth can-i create deployment -n app-team1 --a
yes
student@kube4sure:~$ kubectl auth can-i create deployment -n default --as sy
no
```

CKA-EXAM Alex Pham

2/18 Check

Question 2

Task weight: 3%

Set configuration context:

```
student@kube4sure:~$ kubectl config use-context ek8s
```

Context:

Set the node named ek8s-worker as unavailable and reschedule all the pods running on it.

Hint 💡

Use the kubectl drain comman to drain node.

Solution 🔥

Check status all the nodes in the cluster

```
student@kube4sure:~$ kubectl get nodes
NAME              STATUS  ROLES          AGE  VERSION
ek8s-control-plane  Ready   control-plane  26m  v1.26.6
ek8s-worker         Ready   <none>         26m  v1.26.6
```

Use the kubectl drain command to drain node in the cluster

```
student@kube4sure:~$ kubectl drain ek8s-worker --ignore-daemonsets --delet
```

Check status all the nodes again

```
student@kube4sure:~$ kubectl get nodes
NAME              STATUS                      ROLES          AGE  VERSION
ek8s-control-plane  Ready                       control-plane  27m  v1.26.6
ek8s-worker         Ready,SchedulingDisabled   <none>         27m  v1.26.6
```

Các khái niệm liên quan

Lệnh `kubectl drain` sẽ:

1. **Mark node as unschedulable** (cordon)
2. **Evict pods**
3. **Scheduler reschedule pods sang node khác**

Flow thực tế:

```
kubectl drain node
      ↓
Node SchedulingDisabled
      ↓
Pods bị evict
      ↓
Scheduler tạo pod mới trên node khác
```

Q3. Section: ⚙️ Cluster Maintenance (Upgrade Kubernetes cluster version)

CKA-EXAM Alex Pham

3/18 Check

Question 3

Task weight: 7%

Set configuration context:

```
student@kube4sure:~$ kubectl config use-context mk8s
```

Context:

Given an existing Kubernetes cluster running version 1.30.x, upgrade all of the Kubernetes control plane and node components on the master node only to version 1.31.x.

Be sure to drain the master node before upgrading it and uncordon it after the upgrade.

> You can ssh to the master node using the commande below (password ssh for user student is student)

```
student@kube4sure:~$ ssh mk8s-control-plane
```

> You can assume elevated privileges on the master node with the following command:

```
student@kube4sure:~$ sudo -i
```

You are also expected to upgrade kubelet and kubectl on the master node.

> Do not upgrade worker nodes, etcd, the container manager, the CNI plugin, the DNS service or any other addons.

Hint 💡

Refer to https://kubernetes.io/docs/tasks/administer-cluster/kubeadm/kubeadm-upgrade/

3/18 Check

Refer to https://kubernetes.io/docs/tasks/administer-cluster/kubeadm/kubeadm-upgrade/

Solution 🔥

Check status all the nodes in the cluster

```
student@kube4sure:~$ kubectl get nodes
NAME                STATUS  ROLES          AGE  VERSION
mk8s-control-plane  Ready   control-plane  31m  v1.30.10
mk8s-worker         Ready   <none>         30m  v1.30.10
```

Use the kubectl drain command to drain node in the cluster

```
student@kube4sure:~$ kubectl drain mk8s-control-plane --ignore-daemonsets
```

Check status all the nodes again

```
student@kube4sure:~$ kubectl get nodes
NAME                STATUS                    ROLES          AGE  VERSION
mk8s-control-plane  Ready,SchedulingDisabled  control-plane  37m  v1.30.10
mk8s-worker         Ready                     <none>         36m  v1.30.10
```

SSH into mater node and swith root user

```
student@kube4sure:~$ ssh mk8s-control-plane
student@mk8s-control-plane:~$ sudo -i
root@mk8s-control-plane:~#
```

Unhold Kubernetes package version

```
root@mk8s-control-plane:~# apt update
root@mk8s-control-plane:~# apt-mark unhold kubelet kubeadm kubectl
```

Upgrade kubelet kubeadm kubectl

3/18 Check

Upgrade kubelet kubeadm kubectl

```
root@mk8s-control-plane:~# apt-get install -y kubelet=1.31.* kubeadm=1.31.* k
```

Plan to show stable version

```
root@mk8s-control-plane:~# kubeadm upgrade plan --ignore-preflight-errors=C
```

```
root@mk8s-control-plane:~# kubeadm upgrade apply v1.31.7 --ignore-preflight-
```

Hold Kubernetes package version

```
root@mk8s-control-plane:~# apt-mark hold kubelet kubeadm kubectl
```

Restart Kubelet service

```
root@mk8s-control-plane:~# systemctl daemon-reload
root@mk8s-control-plane:~# systemctl restart kubelet
```

Bring the node back online by marking it schedulable:

```
root@mk8s-control-plane:~# exit
student@mk8s-control-plane:~# exit
student@kube4sure:~$
```

```
student@kube4sure:~# kubectl uncordon mk8s-control-plane
node/mk8s-control-plane uncordoned
```

Check status all the nodes again

```
student@kube4sure:~# kubectl get nodes
NAME                STATUS  ROLES          AGE  VERSION
mk8s-control-plane  Ready   control-plane  85m  v1.31.7
mk8s-worker         Ready   <none>         85m  v1.30.10
```

Q4. etcd operations (backup & restore)

CKA-EXAM **Alex Pham**

4/18 **Check**

Question 4

Task weight: 7%

Set configuration context:

```
student@kube4sure:~$ kubectl config use-context k8s
```

Context:

First, create a snapshot of the existing etcd instance running at https://127.0.0.1:2379 , saving the snapshot to /var/lib/backup/etcd-snapshot.db

> The following TLS certificates/key are supplied for connecting to the server with etcdctl:

- CA certificate: /opt/KUIN00601/ca.crt
- Client certificates: /opt/KUIN00601/etcd-client.crt
- Client key: /opt/KUIN00601/etcd-client.key

> Creating the snapshot of the given instance is expected to complete in seconds.
> If the operation seems to hang, somethings's likely wrong with your command. Use Ctrl+C to cancel the operation and try again.

Next, restore an existing, previous snapshot located at /var/lib/backup/etcd-snapshot-previous.db. To restore this snapshot directly into the running etcd server, switch to etcd user using password: etcd

Hint 💡

Refer to https://kubernetes.io/docs/tasks/administer-cluster/configure-upgrade-etcd/

4/18 **Check**

Question 4

Hint 💡

Solution 🔥

Create a snapshot of the existing etcd instance running

```
student@kube4sure:~$ ETCDCTL_API=3 etcdctl --endpoints https://127.0.0.1:2
  --cert=/opt/KUIN00601/etcd-client.crt \
  --key=/opt/KUIN00601/etcd-client.key \
  --cacert=/opt/KUIN00601/ca.crt \
  snapshot save /var/lib/backup/etcd-snapshot.db
Snapshot saved at /var/lib/backup/etcd-snapshot.db
```

Verify the snapshot:

```
student@kube4sure:~# ETCDCTL_API=3 etcdctl --write-out=table snapshot stat
+----------+----------+------------+------------+
|   HASH   | REVISION | TOTAL KEYS | TOTAL SIZE |
+----------+----------+------------+------------+
| 6d58ef20 |       21 |         24 |      33 kB |
+----------+----------+------------+------------+
```

Before starting the restore operation, a snapshot file must be present

```
student@kube4sure:~# ls /var/lib/backup/
etcd-snapshot.db  etcd-snapshot-previous.db
```

Restoring an etcd cluster

```
student@kube4sure:~# su - etcd
etcd@kube4sure:~# rm -rf /var/lib/etcd/default/
etcd@kube4sure:~# ETCDCTL_API=3 etcdctl  snapshot restore /var/lib/backup/e
etcd@kube4sure:~# sudo systemctl restart etcd
```

Q5. Networking (NetworkPolicy)

CKA-EXAM Alex Pham

< 5/18 > **Check**

Question 5

Task weight: 7%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Create a new NetworkPolicy named allow-port-from-namespace in the existing namespace fubar.

Ensure that the new NetworkPolicy allows Pods in namespace internal to connect to port **9000** of Pods in namespace fubar.

Further ensure that the new NetworkPolicy:

- does not allow access to Pods, which don't listen on port **9000**
- does not allow access from Pods, which are not in namespace internal

Hint 💡

Firstly, we need to set label for namespace. Then, we only create Network Policy which matched the label we created before.

CKA-EXAM Alex Pham

< 5/18 > **Check**

Question 5

Hint 💡

Solution 🔥

Create the Network Policy as following below:

```
student@kube4sure:~# vim np.yaml
```

np.yaml

```yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: allow-port-from-namespace
  namespace: fubar
spec:
  podSelector:
  policyTypes:
    - Ingress
  ingress:
    - from:
        - namespaceSelector:
            matchLabels:
              kubernetes.io/metadata.name: internal
      ports:
        - protocol: TCP
          port: 9000
```

```
student@kube4sure:~# kubectl create -f np.yaml
```

Q6. Services & Networking (Service exposure)

CKA-EXAM Alex Pham

6/18 Check

Question 6

Task weight: 7%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Reconfigure the existing deployment front-end and add a port specification named http exposing port *80/tcp* of the existing container nginx.

Create a new service named front-end-svc exposing the container port http.

Configure the new service to also expose the individual Pods via a NodePort on the nodes on which they are scheduled.

Hint 💡

Use the command kubectl get to get deployment existing and kubectl edit to edit that deployment. Then, use the command kubectl expose to expose pod via service.

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6/18 Check

Question 6

Hint 💡

Solution 🔥

Edit the deployment existing with named front-end and add the port specification under container section like below.

```
student@kube4sure:~# kubectl edit deployment front-end
```

```
...
  spec:
    containers:
    - image: nginx:1.14.2
      imagePullPolicy: IfNotPresent
      name: nginx
      ports:
      - containerPort: 80
        name: http
      resources: {}
      terminationMessagePath: /dev/termination-log
      terminationMessagePolicy: File
...
```

Expose that pod

```
student@kube4sure:~# kubectl expose deployment front-end –name=front-end
```

To get all the infromation of service

```
student@kube4sure:~# kubectl describe svc front-end-svc
```

Q7. Workloads & Scheduling (Scale the deployment presentation to 3 pods)

CKA-EXAM Alex Pham

7/18 Check

Question 7

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Scale the deployment presentation to 3 pods.

Hint 💡

To scale the deployment, please use the `kubectl scale` command.

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7/18 Check

Question 7

Hint 💡

Solution 🔥

Use the `kubectl get` command to list all deployment existing

```
student@kube4sure:~# kubectl get deployment
```

Scale the deployment

```
student@kube4sure:~# kubectl scale deployment presentation –replicas=3
```

Verify the changes

```
student@kube4sure:~# kubectl get deployment
```

Q8. Workloads & Scheduling (Pod scheduling constraints)

CKA-EXAM Alex Pham

8/18 Check

Question 8

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Schedule a pod as follows:

- Name: nginx-kusc00401
- Image: nginx
- Node selector: disk=ssd

Hint 💡

Create a new Pod template which can be used the kubectl run command.

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8/18 Check

Question 8

Hint 💡

Solution 🔥

Create a file pod.yaml

```
student@kube4sure:~# vim pod.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-kusc00401
spec:
  containers:
  - image: nginx
    name: nginx-kusc00401
  nodeSelector:
    disk: ssd
```

Create a Pod with nodeSelector

```
student@kube4sure:~$ kubectl create -f pod.yaml
```

Verify again to make sure the Pod provisioned k8s-worker2 correctly

```
student@kube4sure:~$ kubectl get pods -owide | grep 401
nginx-kusc00401          1/1     Running   0          61s   10.244.1.14   k8s-worker
```

Q9. Cluster Architecture, Installation & Configuration

CKA-EXAM Alex Pham

9/18 Check

Question 9

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Check to see how many nodes are ready (not including nodes tainted NoSchedule) and write the number to /opt/KUSC00402/kusc00402.txt.

Hint 💡

Use the command `kubectl get nodes`

Solution 🔥

Use the `kubectl get nodes` command to see all the node in the cluster

```
student@kube4sure:~# kubectl get nodes
```

```
student@kube4sure:~$ kubectl get nodes
NAME        STATUS  ROLES          AGE  VERSION
k8s-master  Ready   control-plane  18h  v1.26.5
k8s-node-1  Ready   <none>         18h  v1.26.5
k8s-node-2  Ready   <none>         18h  v1.26.5
```

```
student@kube4sure:~$ echo "2" > /opt/KUSC00402/kusc00402.txt
student@kube4sure:~$ cat /opt/KUSC00402/kusc00402.txt
2
```

Q10. Workloads & Scheduling (multi-container Pod)

CKA-EXAM Alex Pham

10/18 Check

Question 10

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Schedule a Pod as follows:

Name: multipod

App Containers: 2

Container Name/Images:

- nginx
- consul

Hint 💡

Create a new Pod template which can be used the kubectl run command. Then, please use the kubectl apply to apply the Pod template.

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10/18 Check

Question 10

Hint 💡

Solution 🔥

Create app.yaml which contains the definition of multipod

```
student@kube4sure:~# vim app.yaml
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  creationTimestamp: null
  labels:
    run: multipod
  name: multipod
spec:
  containers:
  - image: consul
    name: consul
  - image: nginx
    name: nginx
```

Use the kubectl apply command to apply that template

```
student@kube4sure:~# kubectl apply -f app.yaml
```

User the kubectl get pod to see all the pod's running

```
student@kube4sure:~# kubectl get pods
```

Q11. Storage - PV,PVC

CKA-EXAM Alex Pham

11/18 Check

Question 11

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context mk8s
```

Context:

Create a persistent volume with name app-data, of capacity 2Gi and access mode ReadOnlyMany. The type of volume is hostPath and its location is /srv/app-data

Hint 💡

Create a new Persistent Volume for this case. Links https://kubernetes.io/docs/tasks/configure-pod-container/configure-persistent-volume-storage/#create-a-persistentvolume

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11/18 Check

Question 11

Hint 💡

Solution 🔥

Use the vim command to create a new yaml file

```
student@kube4sure:~# vim pv.yaml
```

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: app-data
spec:
  capacity:
    storage: 2Gi
  accessModes:
    - ReadOnlyMany
  hostPath:
    path: "/srv/app-data"
```

Create a new Persisten Volume

```
student@kube4sure:~$ kubectl create -f pv.yaml
```

Verify again

```
student@kube4sure:~$ kubectl get pv
NAME      CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS     CLAIM  S
app-data  2Gi       ROX           Retain          Available         32s
```

Q12. Logging & Monitoring

CKA-EXAM **Alex Pham**

12/18 **Check**

Question 12

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Monitor the logs of pod foo and extract log lines corresponding to error file-not-found.

Write them to /opt/KUTR00101/foo.

Hint 💡

Use the kubectl logs to get all the logs

Solution 🔥

Use the kubectl get command to list all of Pod's running

```
student@kube4sure:~# kubectl get pods
```

Use the kubectl logs command to get all the logs of Pod

```
student@kube4sure:~# kubectl logs foo
```

Get the row correctly as the string "file-not-found" and write it to /opt/KUTR00101/foo.

```
student@kube4sure:~# kubectl logs foo | grep "file-not-found" > /opt/KUTR001
```

Q13. Workloads & Scheduling (Đây là multi-container pod pattern (sidecar logging) → thuộc Workloads)

CKA-EXAM Alex Pham

13/18 Check

Question 13

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

An existing Pod needs to be integrated into the Kubernetes built-in logging architecture (e.g. kubectl logs). Adding a streaming sidecar container is a good and common way to accomplish this requirement.

Add a sidecar container named sidecar, using the busybox image, to the existing Pod big-corp-app. The new sidecar container has to run the following command:

```
/bin/sh -c "tail -n+1 -f /var/log/big-corp-app.log
```

Use a Volume, mounted at /var/log, to make the log file big-corp-app.log available to the sidecar container.

Don't modify the specification of the existing container other than adding the required volume mount.

Hint 💡

CKA-EXAM Alex Pham

13/18 Check

Solution 🔥

Create a yaml file like below named pod13.yaml

```
student@kube4sure:~$ vim pod13.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: big-corp-app
spec:
  containers:
  - image: lfcert/monitor:latest
    name: big-corp-app
    env:
    - name: LOG_FILENAME
      value: /var/log/big-corp-app.log
    volumeMounts:
    - name: logs
      mountPath: /var/log
  - name: busybox-container
    image: busybox
    command: ["/bin/sh", "-c"]
    args: ["tail -n+1 -f /var/log/big-corp-app.log"]
    volumeMounts:
    - name: logs
      mountPath: /var/log
  volumes:
  - name: logs
    emptyDir: {}
```

Delete the big-corp-app pod and create new pod with the yaml file created above

```
student@kube4sure:~$ kubectl delete pod big-corp-app --force
student@kube4sure:~$ kubectl create -f pod13.yaml
```

Please wait until the pod is in Running state

```
student@kube4sure:~$ kubectl get pod big-corp-app -owide
NAME          READY  STATUS   RESTARTS  AGE  IP          NODE        NOMINATEI
big-corp-app  2/2    Running  0         13s  10.244.2.9  k8s-worker  <none>
```

Q14. Cluster Architecture, Installation & Configuration (Tìm Pod có cpu usage nhiều nhất)

CKA-EXAM Alex Pham

14/18 Check

Question 14

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

From the pod label name=overloaded-cpu, find pods running high CPU workloads and write the name of the pod consuming most CPU to the file /opt/KUTR00401/KUTR00401.txt (which already exists).

Hint 💡

Use the kubectl top to get

Solution 🔥

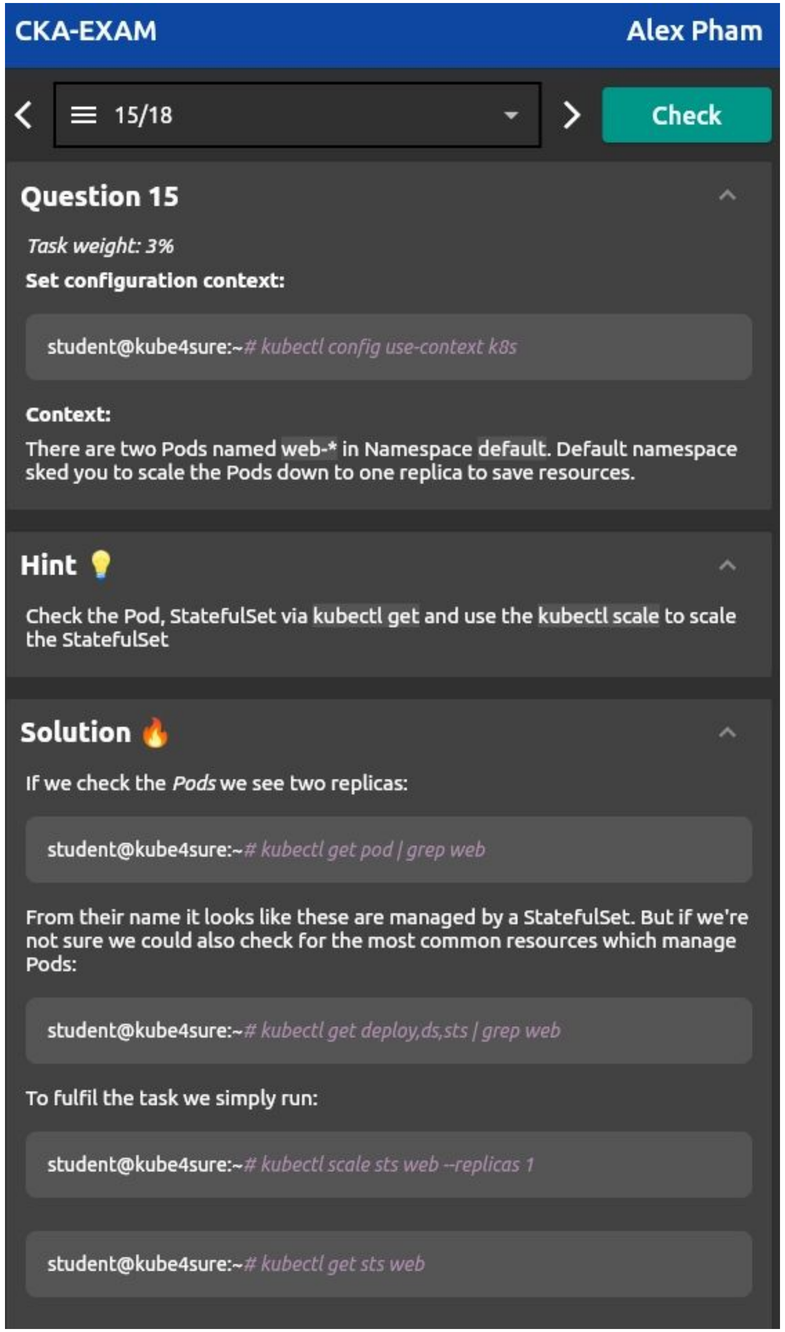
Use the kubectl top command

```
student@kube4sure:~# kubectl top pods
```

```
student@kube4sure:~# kubectl top pods -l name=overloaded-cpu --sort-by=cpu
```

```
--sort-by=cpu | grep overload | head -n 1 > /opt/KUTR00401/KUTR00401.txt
```

Q15. Workloads & Scheduling (scale StatefulSet)



CKA-EXAM Alex Pham

15/18 Check

Question 15

Task weight: 3%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

There are two Pods named web-* in Namespace default. Default namespace sked you to scale the Pods down to one replica to save resources.

Hint 💡

Check the Pod, StatefulSet via kubectl get and use the kubectl scale to scale the StatefulSet

Solution 🔥

If we check the *Pods* we see two replicas:

```
student@kube4sure:~# kubectl get pod | grep web
```

From their name it looks like these are managed by a StatefulSet. But if we're not sure we could also check for the most common resources which manage Pods:

```
student@kube4sure:~# kubectl get deploy,ds,sts | grep web
```

To fulfil the task we simply run:

```
student@kube4sure:~# kubectl scale sts web –replicas 1
```

```
student@kube4sure:~# kubectl get sts web
```

Q16. Storage (PVC, StorageClass)

CKA-EXAM Alex Pham

16/18 Check

Question 16

Task weight: 7%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Create a new PersistentVolumeClaim:

- Name: pv-volume
- Class: csi-hostpath-sc
- Capacity: 10Mi

Create a new Pod which mounts the PersistentVolumeClaim as a volume:

- Name: web-server
- Image: nginx
- Mount path: /usr/share/nginx/html

Configure the new Pod to have ReadWriteOnce access on the volume.

Hint 💡

Create a new Pod which can be used the kubectlcommand.

Create a new Persistent Volume Claim.

CKA-EXAM Alex Pham

16/18 Check

Solution 🔥

Create a pvc.yaml file

```
student@kube4sure:~$ vim pvc.yaml
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: pv-volume
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Mi
  storageClassName: csi-hostpath-sc
```

```
student@kube4sure:~$ kubectl create -f pvc.yaml
```

Create a pod16.yaml file

```
student@kube4sure:~# vim pod16.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: web-server
spec:
  containers:
    - name: web-server
      image: nginx
      volumeMounts:
        - mountPath: "/usr/share/nginx/html"
          name: task-pv-storage
  volumes:
    - name: task-pv-storage
      persistentVolumeClaim:
        claimName: pv-volume
```

Create the pod which will use the PVC

Create the pod which will use the PVC

```
student@kube4sure:~$ kubectl create -f pod16.yaml
```

Please wait until the pod is in Running state

```
student@kube4sure:~$ kubectl get pod web-server -owide
NAME        READY  STATUS   RESTARTS  AGE  IP          NODE         NOMINATED
web-server  1/1    Running  0         44s  10.244.1.9  k8s-worker2  <none>
```

Q17. Services & Networking (Tạo Ingress resource)

CKA-EXAM Alex Pham

17/18 Check

Question 17

Task weight: 7%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context k8s
```

Context:

Create a new nginx Ingress resource as follows:

- Name: pong
- Namespace: ing-internal
- Exposing service hello on path /hello using service port **5678**

The availability of service hello can be checked using the following command, which should return hello:

```
student@kube4sure:~# curl -kL <INTERNAL_IP>/hello
OR
student@kube4sure:~# curl -kL localhost/hello
```

Hint 💡

Create a new Ingress resource that can be followed in the official document of Kubernetes.

https://kubernetes.io/docs/concepts/services-networking/ingress/

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17/18 Check

Question 17

Hint 💡

Solution 🔥

Use the vim editor to create a new Ingress, which should be named ingress-internal.yaml

```
student@kube4sure:~$ vim ingress-internal.yaml
```

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ping
  namespace: ing-internal
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
spec:
  ingressClassName: nginx
  rules:
  - http:
      paths:
      - path: /hello
        pathType: Prefix
        backend:
          service:
            name: hello
            port:
              number: 5678
```

```
student@kube4sure:~$ kubectl create -f ingress-internal.yaml
```

Q18. Cluster Architecture, Installation & Configuration (cluster troubleshooting)

CKA-EXAM Alex Pham

18/18 Check

Question 18

Task weight: 13%

Set configuration context:

```
student@kube4sure:~# kubectl config use-context mk8s
```

Context:

A Kubernetes worker node, named mk8s-worker is in state NotReady. Investigate why this is the case, and perform any appropriate steps to bring the node to a Ready state, ensuring that any changes are made permanent.

You can ssh to the failed node using:

```
student@kube4sure:~# ssh mk8s-worker
```

You can assume elevated privileges on the node with the following command:

```
student@mk8s-worker:~# sudo -i
```

Hint 💡

Check the status of the Kubelet service in the mk8s-worker node.

18/18 Check

Question 18

Hint 💡

Solution 🔥

SSH into worker node

```
student@kube4sure:~$ ssh mk8s-worker
```

Check the status of the Kubelet service in the mk8s-worker node.

```
student@mk8s-worker:~$ systemctl status kubelet
```

Ensuring that any changes are made permanently means starting the Kubelet service automatically after rebooting the node.

Switch to root user

```
student@mk8s-worker:~$ sudo -i
```

Enable and restart the Kubelet service automatically

```
student@mk8s-worker:~# systemctl enable --now kubelet
student@mk8s-worker:~# systemctl restart kubelet
```